

This compiler flag forces the C compiler to use the **C89** standard (also known as ANSI C or C90). [\[1, 2, 3, 4\]](#)

Key Features of C89

- **Oldest standard:** Released in 1989.
- **High compatibility:** Supported by almost every C compiler.
- **Strict rules:** Lacks many modern features. [\[1, 2, 3, 4, 5\]](#)

Missing Features (Not allowed in C89)

- **No `//` comments:** Only `/* ... */` is allowed.
- **Mixed declarations:** Variables must be declared at the top of a block.
- **No inline functions:** The `inline` keyword is not supported.
- **No `restrict` keyword:** This pointer optimization does not exist.
- **No `long long` type:** 64-bit integers are not standard.
- **For-loop declarations:** You cannot declare variables inside a `for` loop statement (e.g., `for (int i = 0; ...)` is invalid). [\[1, 2, 3, 4, 5\]](#)

Common Use Cases

- **Legacy systems:** Maintaining very old source code.
- **Embedded systems:** Writing code for highly restricted microcontrollers.
- **Maximum portability:** Ensuring code compiles on any system. [\[1, 2, 3, 4, 5\]](#)